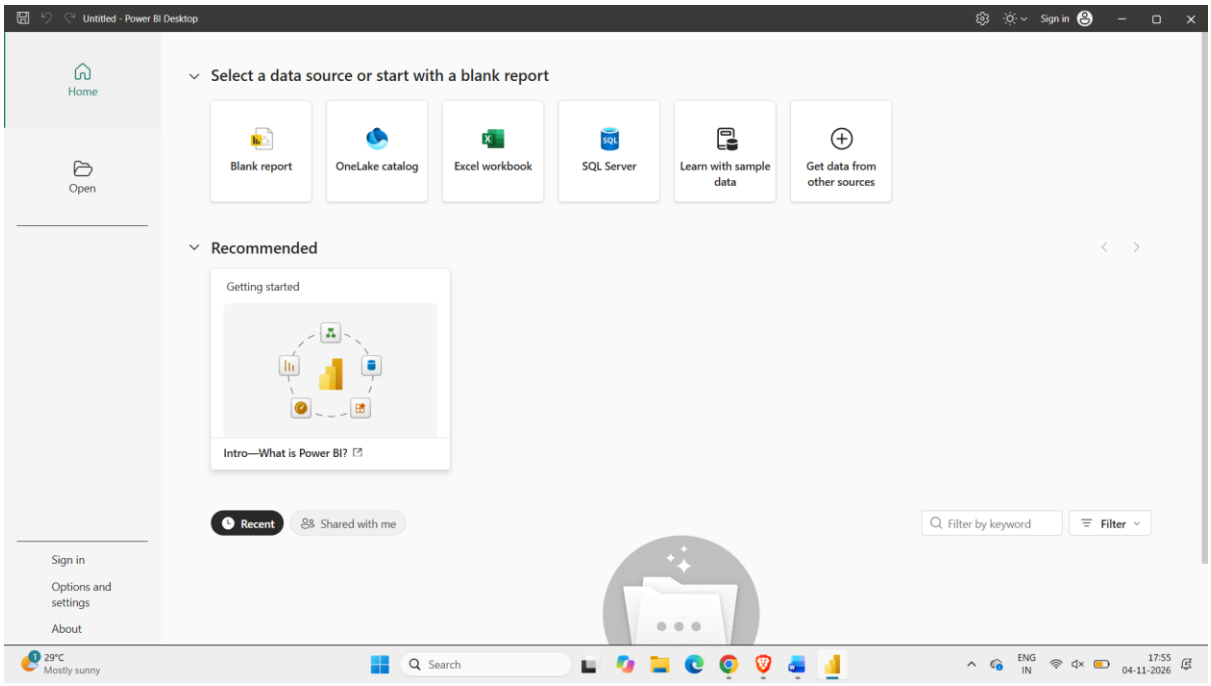
Task – 3


Untitled - Power BI Desktop
Sign in
Home
Open
Select a data source or start with a blank report
Blank report
OneLake catalog
Excel workbook
SQL Server
Learn with sample data
Get data from other sources
Recommended
Getting started
Intro—What is Power BI?
Recent
Shared with me
Filter by keyword
Filter
Sign in
Options and settings
About
29°C
Mostly sunny
Search
ENG
IN
17:55
04-11-2026

ELECTRICITY_BILLS.csv

ELECTRICITY_BILLS.csv

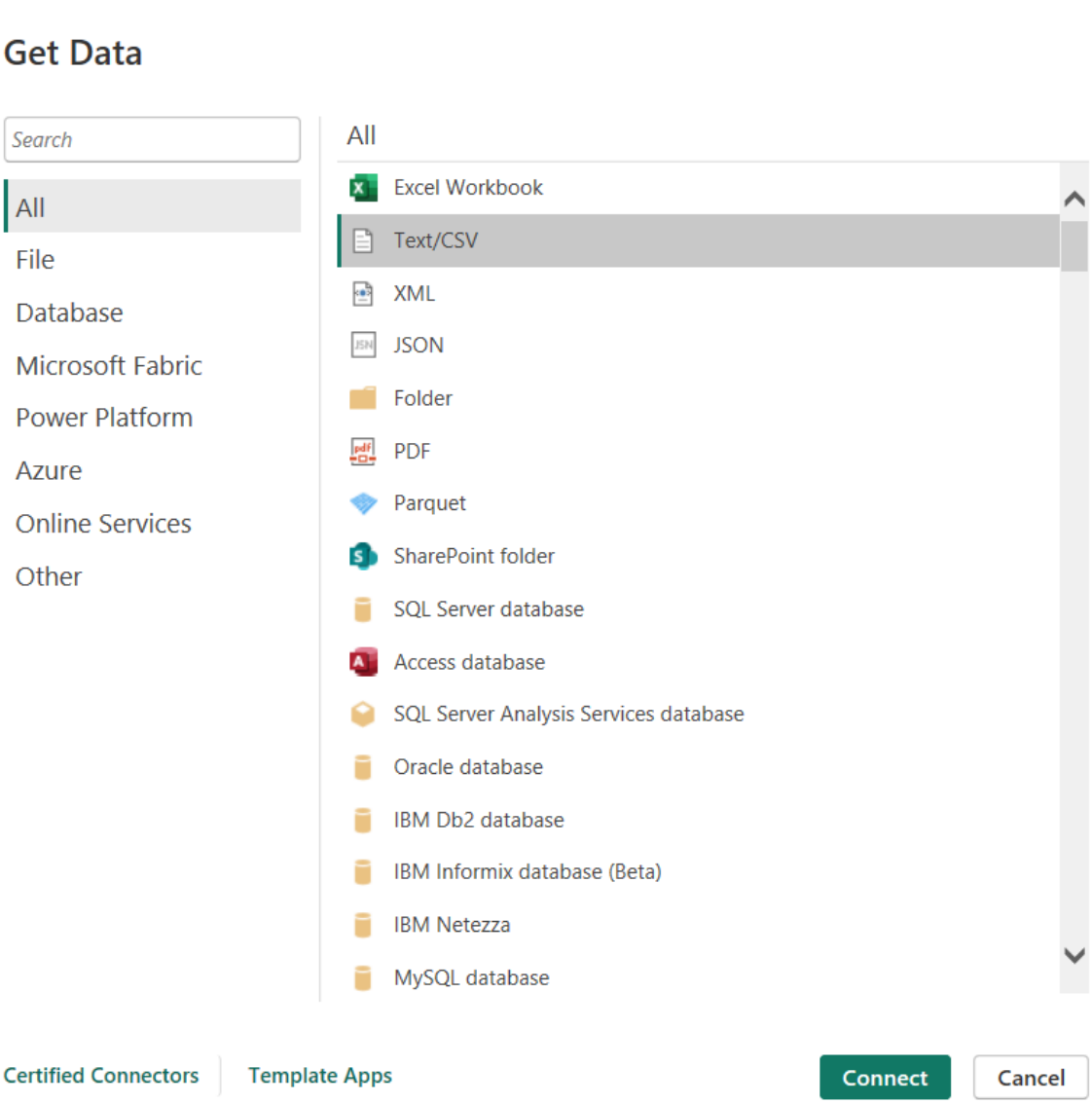
```
BILL_ID,CUSTOMER_ID,BILLING_MONTH,UNITS_CONSUMED,RATE_PER_UNIT,SURCHARGE,BILL_AMOUNT
B1001,C001,2026-01-01,250,5.5,20,1395.0
B1002,C002,2026-01-01,1200,8.2,150,9990.0
B1003,C003,2026-01-01,5000,7.5,500,38000.0
B1004,C004,2026-01-01,180,5.5,15,1005.0
B1005,C005,2026-02-01,320,8.0,60,2620.0
B1006,C006,2026-02-01,450,5.8,25,2635.0
B1007,C007,2026-02-01,900,4.2,0,3780.0
B1008,C008,2026-02-01,275,5.6,18,1558.0
B1009,C009,2026-02-01,1500,8.1,120,12270.0
B1010,C010,2026-02-01,210,5.5,15,1170.0
B1011,C011,2026-02-01,4200,7.4,400,31480.0
B1012,C012,2026-02-01,1100,8.3,100,9230.0
B1013,C013,2026-03-01,300,5.7,20,1730.0
B1014,C014,2026-03-01,750,4.1,0,3075.0
B1015,C015,2026-03-01,1800,8.0,160,14560.0
B1016,C016,2026-03-01,380,5.8,20,2224.0
B1017,C017,2026-03-01,1350,8.2,110,11180.0
B1018,C018,2026-03-01,4800,7.6,450,36930.0
B1019,C019,2026-03-01,290,5.6,18,1642.0
B1020,C020,2026-03-01,680,4.3,0,2924.0
B1021,C021,2026-04-01,410,5.7,25,2362.0
B1022,C022,2026-04-01,1250,8.1,100,10225.0
B1023,C023,2026-04-01,260,5.6,15,1471.0
B1024,C024,2026-04-01,1600,8.0,130,12930.0
B1025,C001,2026-02-01,280,5.5,20,1560.0
B1026,C002,2026-03-01,1350,8.2,150,11220.0
B1027,C003,2026-03-01,5300,7.5,520,40270.0
B1028,C006,2026-03-01,470,5.8,25,2751.0
B1029,C009,2026-03-01,1420,8.1,115,11617.0
B1030,C011,2026-04-01,3900,7.4,380,29240.0
B1031,C014,2026-04-01,820,4.1,0,3362.0
B1032,C017,2026-04-01,1280,8.2,105,10601.0
B1033,C018,2026-04-01,5100,7.6,460,39220.0
B1034,C024,2026-04-01,1720,8.0,135,13895.0
```

ELECTRICITY_CUSTOMERS.csv

ELECTRICITY_CUSTOMERS.csv

```csv
CUSTOMER_ID,CUSTOMER_NAME,CUSTOMER_TYPE,CITY,CONNECTION_TYPE,EMAIL
C001,Arjun Kumar,Residential,Chennai,Single Phase,arjun@example.com
C002,Priya Sharma,Commercial,Bengaluru,Three Phase,priya@example.com
C003,Kiran Reddy,Industrial,Hyderabad,Three Phase,kiran@example.com
C004,Meera Nair,Residential,Kochi,Single Phase,meera@example.com
C005,Vikram Singh,Commercial,Pune,Three Phase,vikram@example.com
C006,Ananya Rao,Residential,Chennai,Single Phase,ananya@example.com
C007,Rahul Das,Agricultural,Vijayawada,Three Phase,rahul@example.com
C008,Sneha Iyer,,Coimbatore,Single Phase,sneha@example.com
C009,Manoj Gupta,Commercial,Mumbai,Three Phase,manoj@example.com
C010,Divya Menon,Residential,Kochi,Single Phase,divya@example.com
C011,Suresh Babu,Industrial,Visakhapatnam,Three Phase,suresh@example.com
C012,Pooja Verma,Commercial,Delhi,Three Phase,pooja@example.com
C013,Nikhil Jain,   ,Jaipur,Single Phase,nikhil@example.com
C014,Lakshmi Devi,Agricultural,Guntur,Three Phase,lakshmi@example.com
C015,Amit Shah,Commercial,Ahmedabad,Three Phase,amit@example.com
C016,Keerthi Raju,Residential,Hyderabad,Single Phase,keerthi@example.com
C017,Farhan Ali,Commercial,Chennai,Three Phase,farhan@example.com
C018,Rohit Patil,Industrial,Pune,Three Phase,rohit@example.com
C019,Sanjana Bose,Residential,Kolkata,Single Phase,sanjana@example.com
C020,Harish Rao,Agricultural,Mysuru,Three Phase,harish@example.com
C021,Neha Kapoor,Residential,Mumbai,Single Phase,neha@example.com
C022,Gopal Krishna,Commercial,Vijayawada,Three Phase,gopal@example.com
C023,Ishita Roy,Residential,Kolkata,Single Phase,ishita@example.com
C024,David Thomas,Commercial,Kochi,Three Phase,david@example.com

```


Get Data
Search
All
File
Database
Microsoft Fabric
Power Platform
Azure
Online Services
Other
All
Excel Workbook
Text/CSV
XML
JSON
Folder
PDF
Parquet
SharePoint folder
SQL Server database
Access database
SQL Server Analysis Services database
Oracle database
IBM Db2 database
IBM Informix database (Beta)
IBM Netezza
MySQL database
Certified Connectors
Template Apps
Connect
Cancel

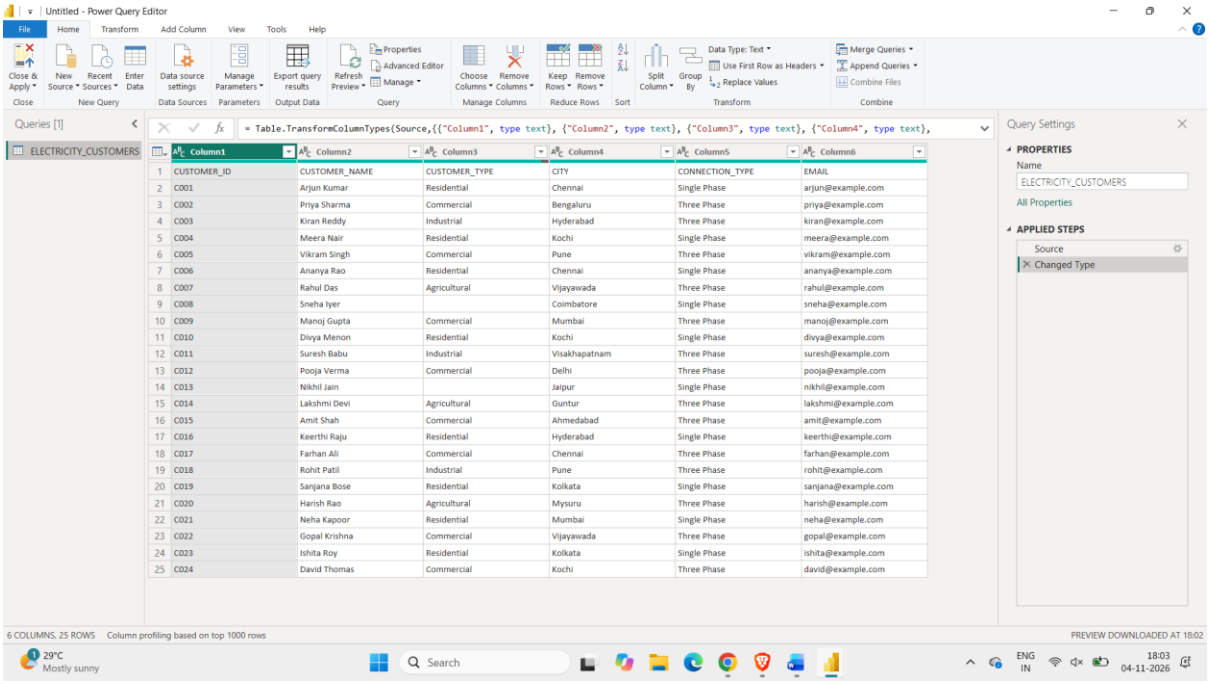
ELECTRICITY_CUSTOMERS.csv

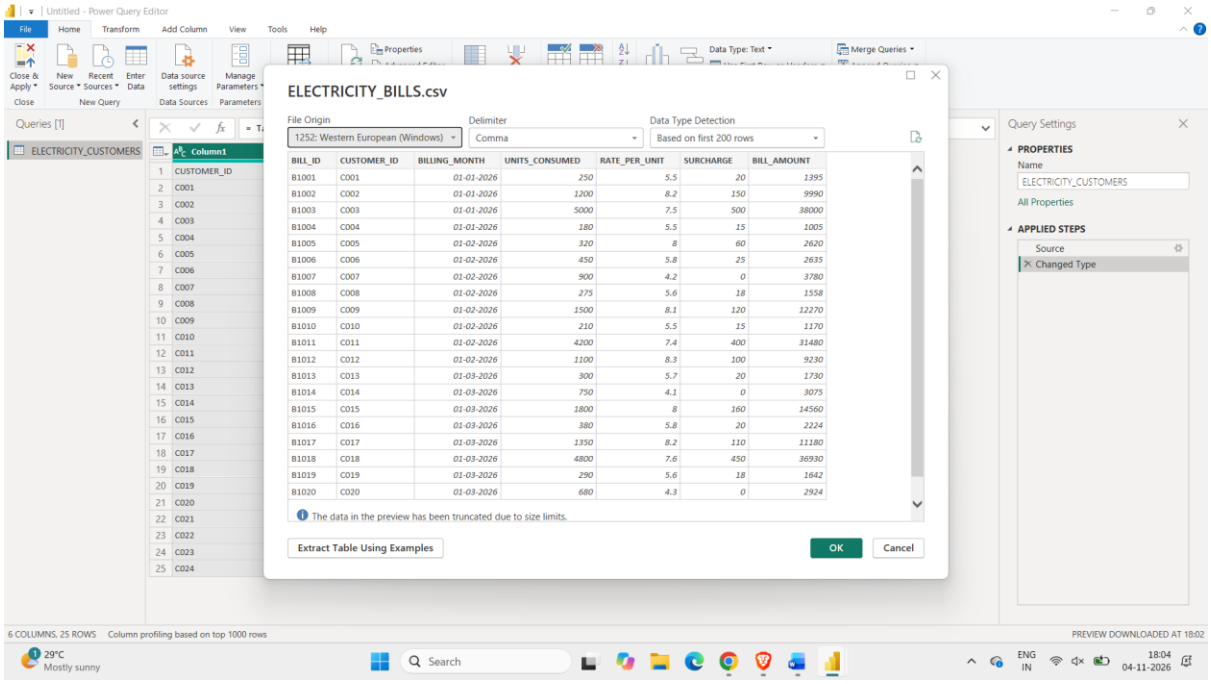
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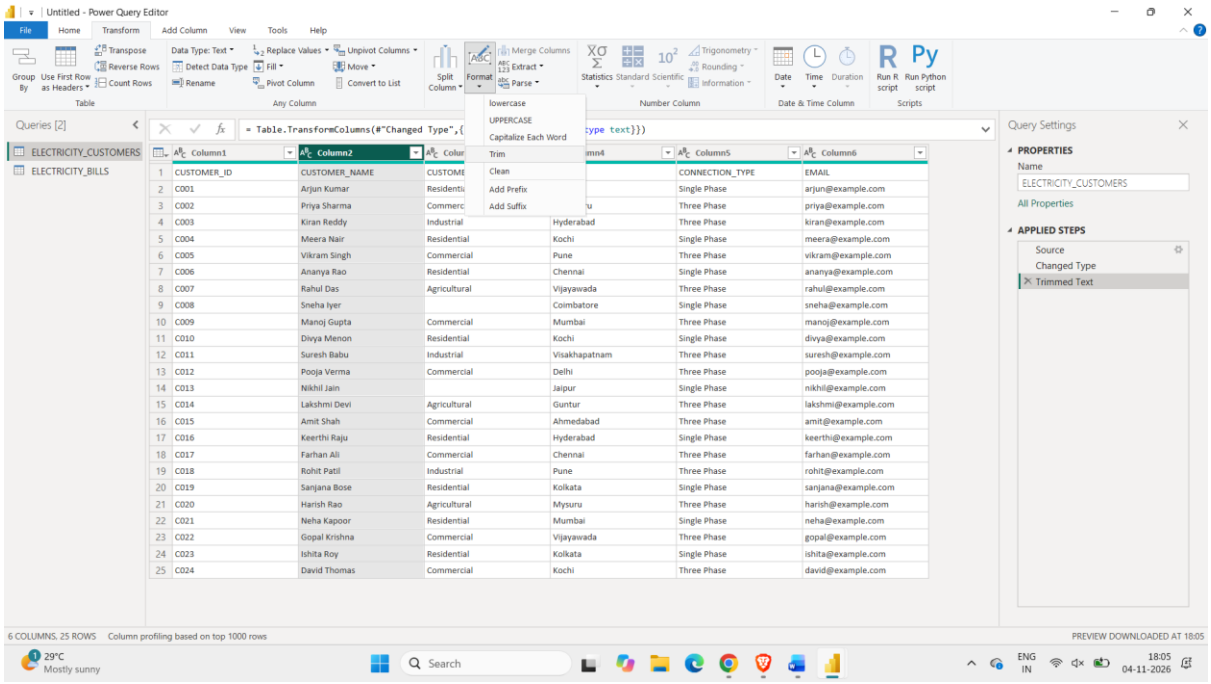
Column1	Column2	Column3	Column4	Column5	Column6
CUSTOMER_ID	CUSTOMER_NAME	CUSTOMER_TYPE	CITY	CONNECTION_TYPE	EMAIL
C001	Arjun Kumar	Residential	Chennai	Single Phase	arjun@example.com
C002	Priya Sharma	Commercial	Bengaluru	Three Phase	priya@example.com
C003	Kiran Reddy	Industrial	Hyderabad	Three Phase	kiran@example.com
C004	Meera Nair	Residential	Kochi	Single Phase	meera@example.com
C005	Vikram Singh	Commercial	Pune	Three Phase	vikram@example.com
C006	Ananya Rao	Residential	Chennai	Single Phase	ananya@example.com
C007	Rahul Das	Agricultural	Vijayawada	Three Phase	rahul@example.com
C008	Sneha Iyer		Coimbatore	Single Phase	sneha@example.com
C009	Manoj Gupta	Commercial	Mumbai	Three Phase	manoj@example.com
C010	Divya Menon	Residential	Kochi	Single Phase	divya@example.com
C011	Suresh Babu	Industrial	Visakhapatnam	Three Phase	suresh@example.com
C012	Pooja Verma	Commercial	Delhi	Three Phase	pooja@example.com
C013	Nikhil Jain		Jaipur	Single Phase	nikhil@example.com
C014	Lakshmi Devi	Agricultural	Guntur	Three Phase	lakshmi@example.com
C015	Amit Shah	Commercial	Ahmedabad	Three Phase	amit@example.com
C016	Keerthi Raju	Residential	Hyderabad	Single Phase	keerthi@example.com
C017	Farhan Ali	Commercial	Chennai	Three Phase	farhan@example.com
C018	Rohit Patil	Industrial	Pune	Three Phase	rohit@example.com
C019	Sanjana Bose	Residential	Kolkata	Single Phase	sanjana@example.com

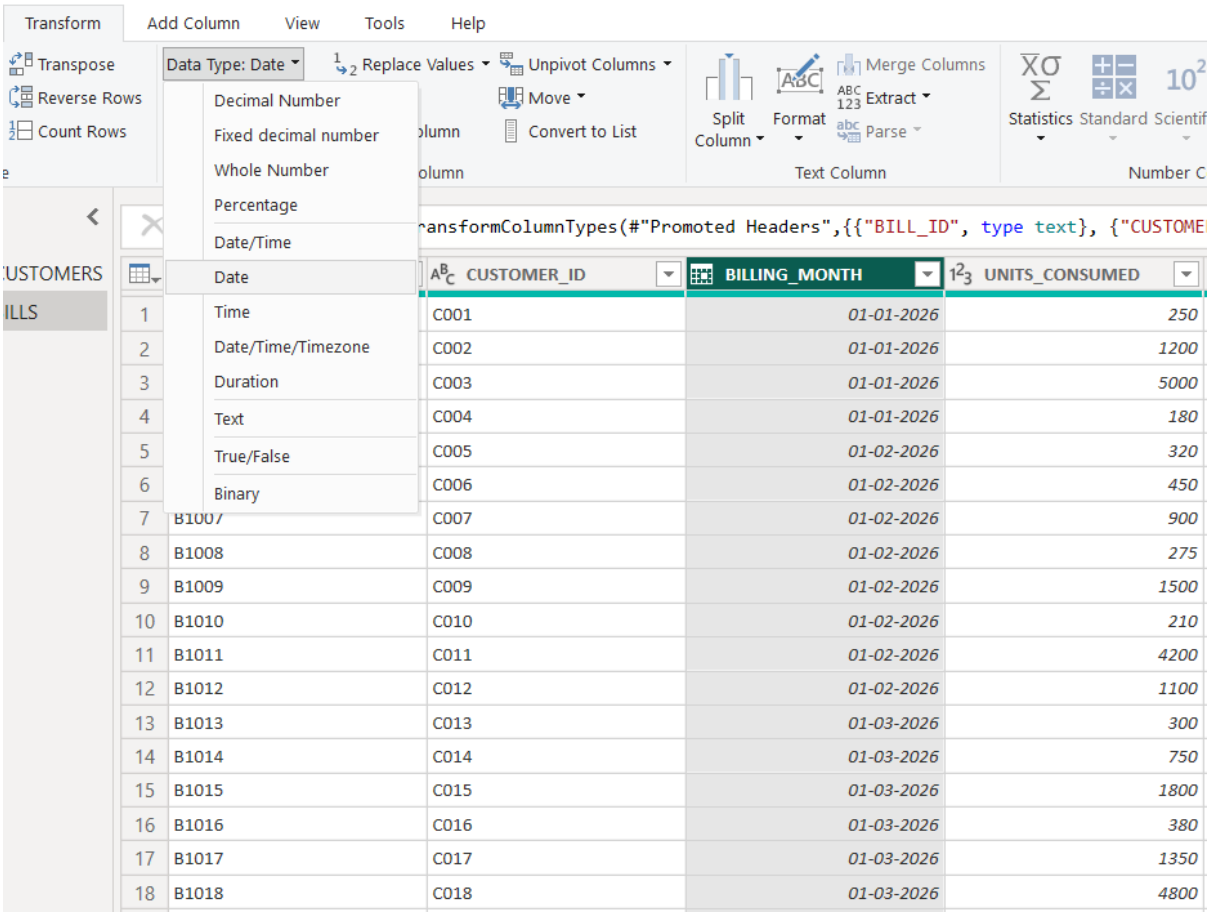
The data in the preview has been truncated due to size limits.

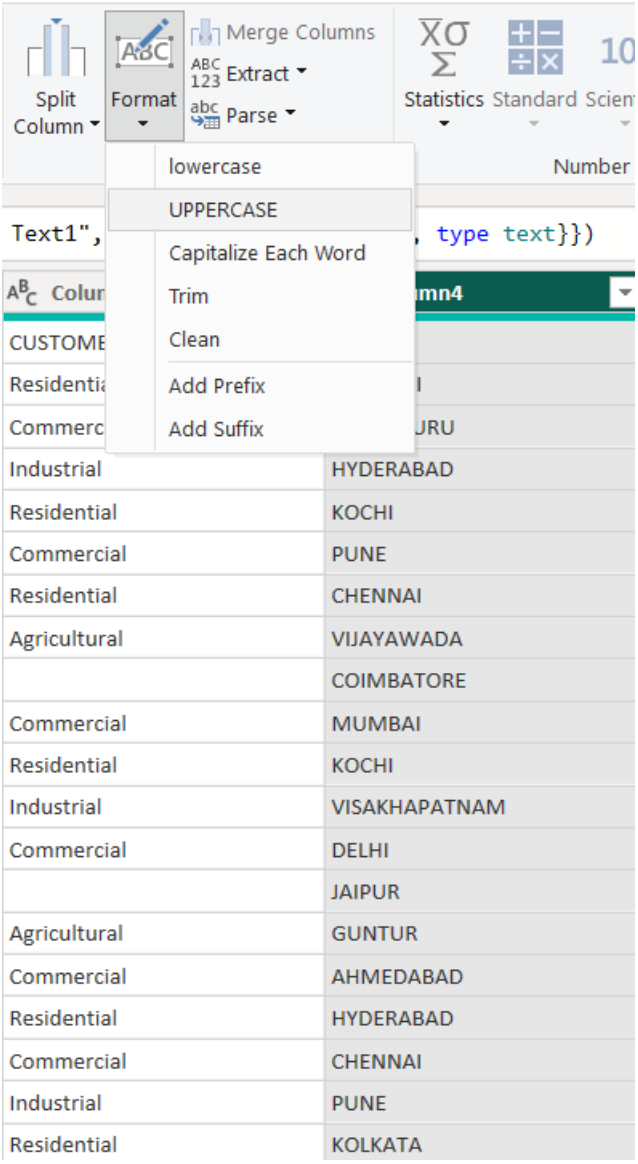
Extract Table Using Examples | Load | Transform Data | Cancel

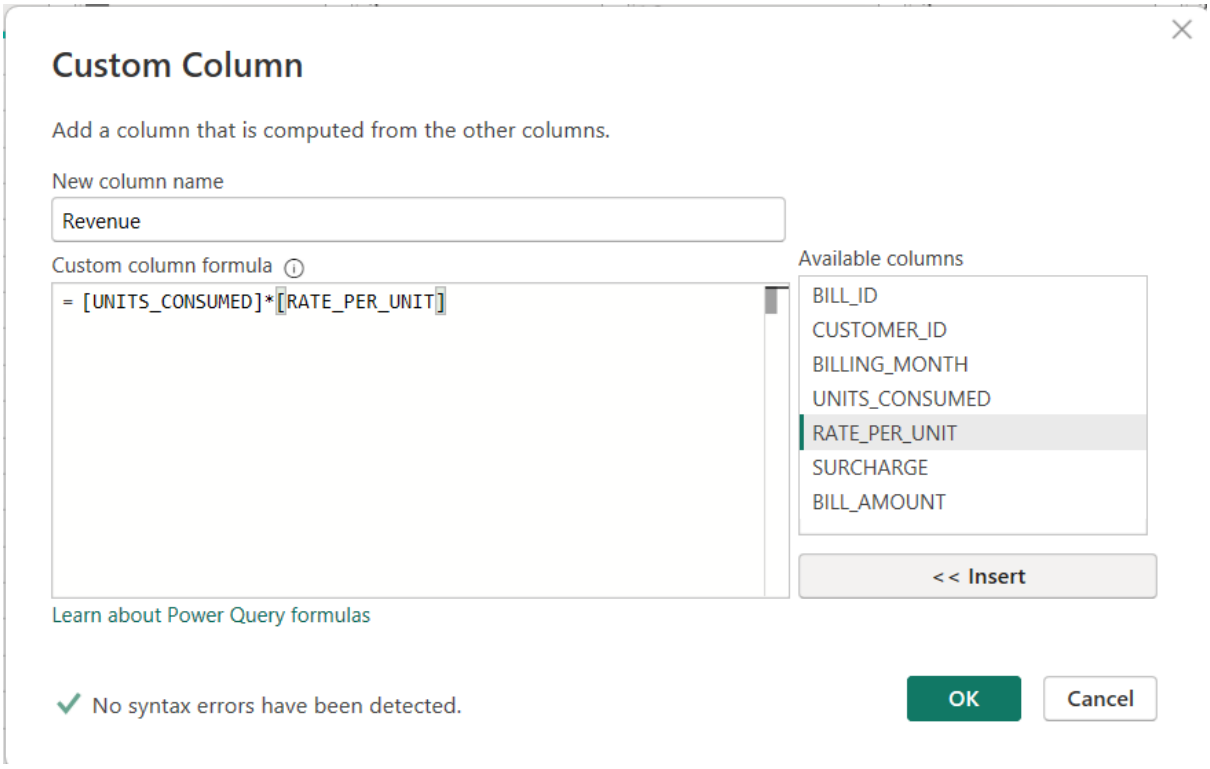



Untitled - Power Query Editor
ELECTRICITY_BILLS.csv
File Origin
1252: Western European (Windows)
Delimiter
Comma
Data Type Detection
Based on first 200 rows
BILL_ID | CUSTOMER_ID | BILLING_MONTH | UNITS_CONSUMED | RATE_PER_UNIT | SURCHARGE | BILL_AMOUNT
B1001 | C001 | 01-01-2026 | 250 | 5.5 | 20 | 1395
B1002 | C002 | 01-01-2026 | 1200 | 8.2 | 150 | 9990
B1003 | C003 | 01-01-2026 | 5000 | 7.5 | 500 | 38000
B1004 | C004 | 01-01-2026 | 180 | 5.5 | 15 | 1005
B1005 | C005 | 01-02-2026 | 320 | 8 | 60 | 2620
B1006 | C006 | 01-02-2026 | 450 | 5.8 | 25 | 2635
B1007 | C007 | 01-02-2026 | 900 | 4.2 | 0 | 3780
B1008 | C008 | 01-02-2026 | 275 | 5.6 | 18 | 1558
B1009 | C009 | 01-02-2026 | 1500 | 8.1 | 120 | 12270
B1010 | C010 | 01-02-2026 | 210 | 5.5 | 15 | 1170
B1011 | C011 | 01-02-2026 | 4200 | 7.4 | 400 | 31480
B1012 | C012 | 01-02-2026 | 1100 | 8.3 | 100 | 9230
B1013 | C013 | 01-03-2026 | 300 | 5.7 | 20 | 1730
B1014 | C014 | 01-03-2026 | 750 | 4.1 | 0 | 3075
B1015 | C015 | 01-03-2026 | 1800 | 8 | 160 | 14560
B1016 | C016 | 01-03-2026 | 380 | 5.8 | 20 | 2224
B1017 | C017 | 01-03-2026 | 1350 | 8.2 | 110 | 11180
B1018 | C018 | 01-03-2026 | 4800 | 7.6 | 450 | 36930
B1019 | C019 | 01-03-2026 | 290 | 5.6 | 18 | 1642
B1020 | C020 | 01-03-2026 | 680 | 4.3 | 0 | 2924
The data in the preview has been truncated due to size limits.
Extract Table Using Examples
OK
Cancel
Queries [1]
ELECTRICITY_CUSTOMERS
Query Settings
PROPERTIES
Name
ELECTRICITY_CUSTOMERS
All Properties
APPLIED STEPS
Source
Changed Type
6 COLUMNS, 25 ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 18:02

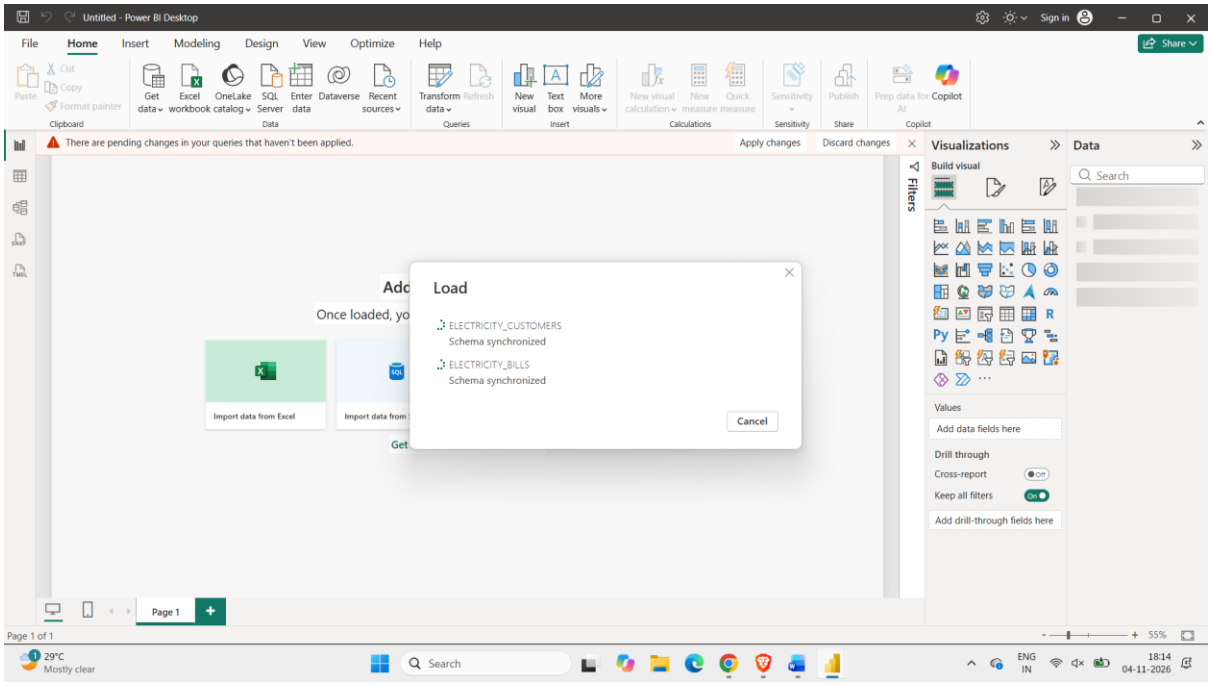

Untitled - Power Query Editor
Queries [2]
ELECTRICITY_CUSTOMERS
ELECTRICITY_BILLS
Format
lowercase
UPPERCASE
Capitalize Each Word
Trim
Clean
Add Prefix
Add Suffix
CUSTOMER_ID | CUSTOMER_NAME | CUSTOME | mn4 | CONNECTION_TYPE | EMAIL
C001 | Arjun Kumar | Residenti | | Single Phase | arjun@example.com
C002 | Priya Sharma | Commerc | u | Three Phase | priya@example.com
C003 | Kiran Reddy | Industrial | Hyderabad | Three Phase | kiran@example.com
C004 | Meera Nair | Residential | Kochi | Single Phase | meera@example.com
C005 | Vikram Singh | Commercial | Pune | Three Phase | vikram@example.com
C006 | Ananya Rao | Residential | Chennai | Single Phase | ananya@example.com
C007 | Rahul Das | Agricultural | Vijayawada | Three Phase | rahul@example.com
C008 | Sneha Iyer | | Coimbatore | Single Phase | sneha@example.com
C009 | Manoj Gupta | Commercial | Mumbai | Three Phase | manoj@example.com
C010 | Divya Menon | Residential | Kochi | Single Phase | divya@example.com
C011 | Suresh Babu | Industrial | Visakhapatnam | Three Phase | suresh@example.com
C012 | Pooja Verma | Commercial | Delhi | Three Phase | pooja@example.com
C013 | Nikhil Jain | | Jaipur | Single Phase | nikhil@example.com
C014 | Lakshmi Devi | Agricultural | Guntur | Three Phase | lakshmi@example.com
C015 | Amit Shah | Commercial | Ahmedabad | Three Phase | amit@example.com
C016 | Keerthi Raju | Residential | Hyderabad | Single Phase | keerthi@example.com
C017 | Farhan Ali | Commercial | Chennai | Three Phase | farhan@example.com
C018 | Rohit Patil | Industrial | Pune | Three Phase | rohit@example.com
C019 | Sanjana Bose | Residential | Kolkata | Single Phase | sanjana@example.com
C020 | Harish Rao | Agricultural | Mysuru | Three Phase | harish@example.com
C021 | Neha Kapoor | Residential | Mumbai | Single Phase | neha@example.com
C022 | Gopal Krishna | Commercial | Vijayawada | Three Phase | gopal@example.com
C023 | Ishita Roy | Residential | Kolkata | Single Phase | ishita@example.com
C024 | David Thomas | Commercial | Kochi | Three Phase | david@example.com
Query Settings
PROPERTIES
Name
ELECTRICITY_CUSTOMERS
All Properties
APPLIED STEPS
Source
Changed Type
Trimmed Text
6 COLUMNS, 25 ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 18:05


Transform
Add Column
View
Tools
Help
Transpose
Reverse Rows
Count Rows
Data Type: Date
Replace Values
Unpivot Columns
Move
Convert to List
Split Column
Format
Merge Columns
Extract
Parse
Text Column
Statistics
Standard
Number C
Decimal Number
Fixed decimal number
Whole Number
Percentage
Date/Time
Date
Time
Date/Time/Timezone
Duration
Text
True/False
Binary
ransformColumnTypes(#"Promoted Headers",{{"BILL_ID", type text}, {"CUSTOME
CUSTOMERS
ILLS
CUSTOMER_ID
BILLING_MONTH
UNITS_CONSUMED
1 B1007 C007 01-02-2026 900
2 B1008 C008 01-02-2026 275
3 B1009 C009 01-02-2026 1500
4 B1010 C010 01-02-2026 210
5 B1011 C011 01-02-2026 4200
6 B1012 C012 01-02-2026 1100
7 B1013 C013 01-03-2026 300
8 B1014 C014 01-03-2026 750
9 B1015 C015 01-03-2026 1800
10 B1016 C016 01-03-2026 380
11 B1017 C017 01-03-2026 1350
12 B1018 C018 01-03-2026 4800


Split Column
Format
Merge Columns
Extract
Parse
Statistics
Standard
Number
lowercase
UPPERCASE
Capitalize Each Word
Trim
Clean
Add Prefix
Add Suffix
Text1",
type text}})
Colur
mn4
CUSTOME
Residentia
Commerc
URU
Industrial HYDERABAD
Residential KOCHI
Commercial PUNE
Residential CHENNAI
Agricultural VIJAYAWADA
COIMBATORE
Commercial MUMBAI
Residential KOCHI
Industrial VISAKHAPATNAM
Commercial DELHI
JAIPUR
Agricultural GUNTUR
Commercial AHMEDABAD
Residential HYDERABAD
Commercial CHENNAI
Industrial PUNE
Residential KOLKATA



BILLING_MONTH	UNITS_CONSUMED	RATE_PER_UNIT	SURCHARGE	BILL_AMOUNT	Revenue
01-01-2026	250	5.5	20	1395	1375
01-01-2026	1200	8.2	150	9990	9840
01-01-2026	5000	7.5	500	38000	37500
01-01-2026	180	5.5	15	1005	990
01-02-2026	320	8	60	2620	2560
01-02-2026	450	5.8	25	2635	2610
01-02-2026	900	4.2	0	3780	3780
01-02-2026	275	5.6	18	1558	1540
01-02-2026	1500	8.1	120	12270	12150
01-02-2026	210	5.5	15	1170	1155
01-02-2026	4200	7.4	400	31480	31080
01-02-2026	1100	8.3	100	9230	9130
01-03-2026	300	5.7	20	1730	1710
01-03-2026	750	4.1	0	3075	3075
01-03-2026	1800	8	160	14560	14400
01-03-2026	380	5.8	20	2224	2204
01-03-2026	1350	8.2	110	11180	11070
01-03-2026	4800	7.6	450	36930	36480
01-03-2026	290	5.6	18	1642	1624
01-03-2026	680	4.3	0	2924	2924
01-04-2026	410	5.7	25	2362	2337
01-04-2026	1250	8.1	100	10225	10125
01-04-2026	260	5.6	15	1471	1456
01-04-2026	1600	8	130	12930	12800
01-02-2026	280	5.5	20	1560	1540
01-03-2026	1350	8.2	150	11220	11070
01-03-2026	5300	7.5	520	40270	39750


Untitled - Power BI Desktop
Sign in
File
Home
Insert
Modeling
Design
View
Optimize
Help
Share
Paste
Cut
Copy
Format painter
Clipboard
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Data
Transform data
Refresh
Queries
New visual
Text box
More visuals
Insert
New visual calculation
New measure
Quick measure
Calculations
Sensitivity
Publish
Share
Prep data for AI
Copilot
There are pending changes in your queries that haven't been applied.
Apply changes
Discard changes
Load
ELECTRICITY_CUSTOMERS
Schema synchronized
ELECTRICITY_BILLS
Schema synchronized
Cancel
Import data from Excel
Filters
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Data
Search
Page 1
Page 1 of 1
55%
29°C
Mostly clear
Search
ENG
IN
18:14
04-11-2026

New relationship

Select tables and columns that are related.

From table

ELECTRICITY_CUSTOMERS

Column1	Column2	Column3	Column4	Column5	Column6
CUSTOMER_ID	CUSTOMER_...	CUSTOMER_T...	CITY	CONNECTIO...	EMAIL
C001	Arjun Kumar	Residential	CHENNAI	Single Phase	arjun@examp...
C002	Priya Sharma	Commercial	BENGALURU	Three Phase	priya@examp...

To table

ELECTRICITY_BILLS

BILL_AMOUNT	BILL_ID	BILLING_MO...	CUSTOMER_ID	RATE_PER_UNIT	Revenue	SURCHAR
1395	B1001	01 January 20...	C001	5.5	1375	20
9990	B1002	01 January 20...	C002	8.2	9840	150
38000	B1003	01 January 20...	C003	7.5	37500	500

Cardinality

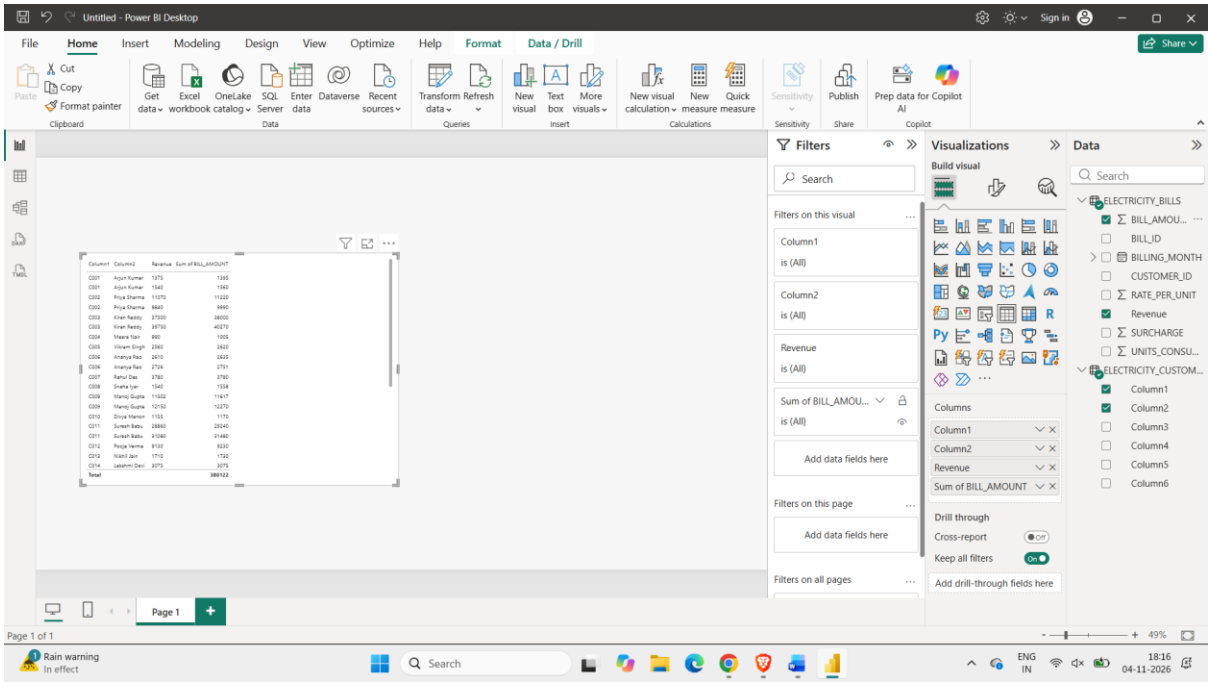
One to many (1:*)

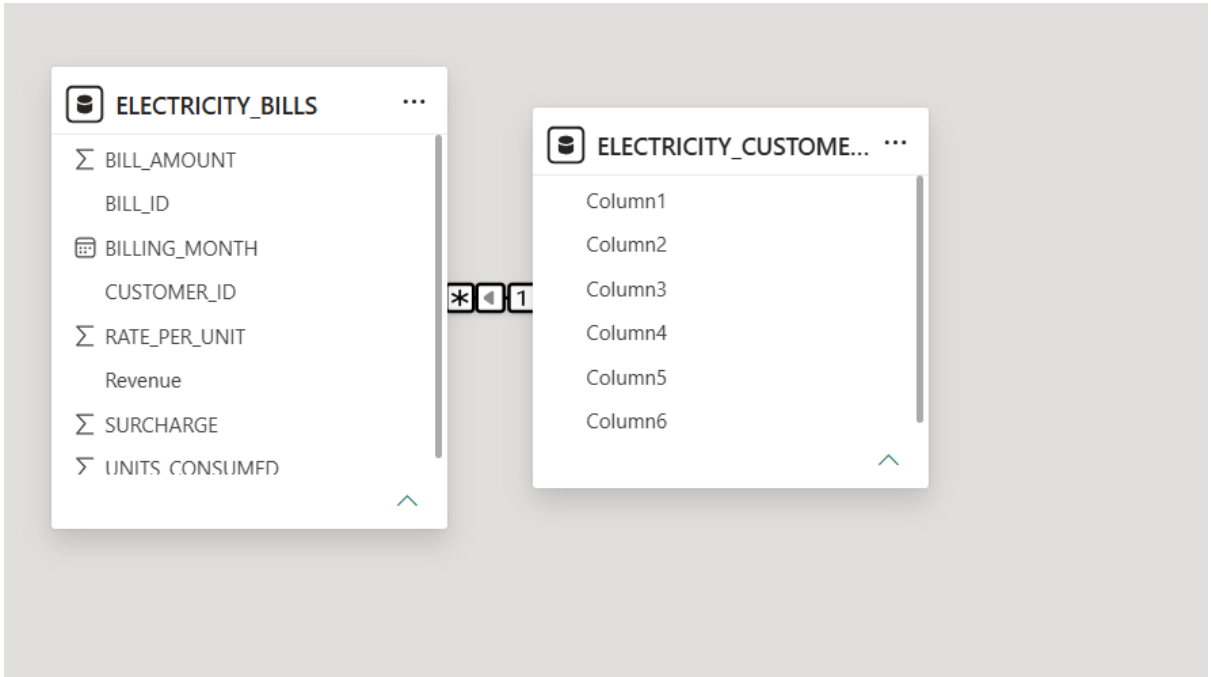
Cross-filter direction

Single

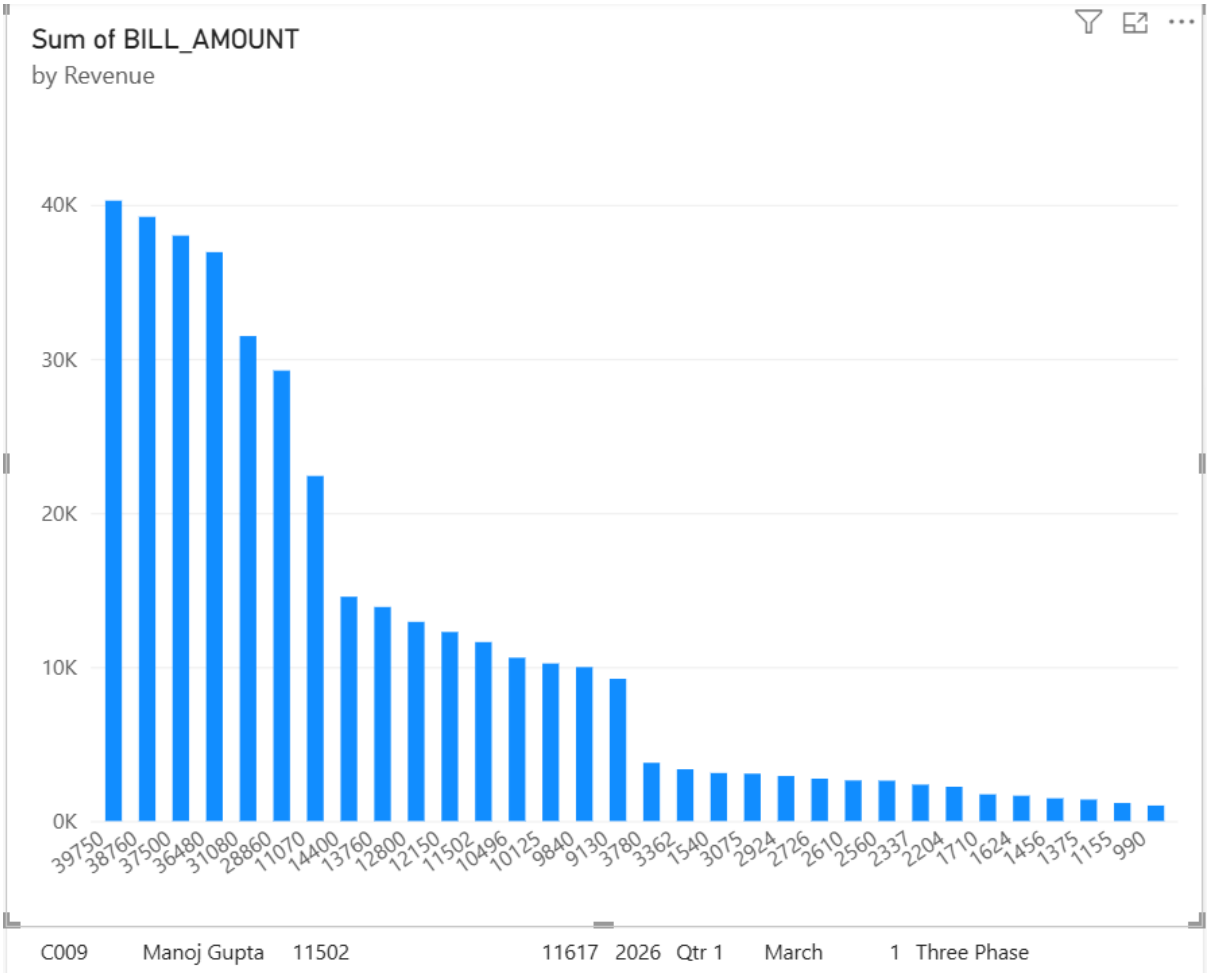
- [x] Make this relationship active
- [ ] Assume referential integrity
- [ ] Apply security filter in both directions

Save Cancel

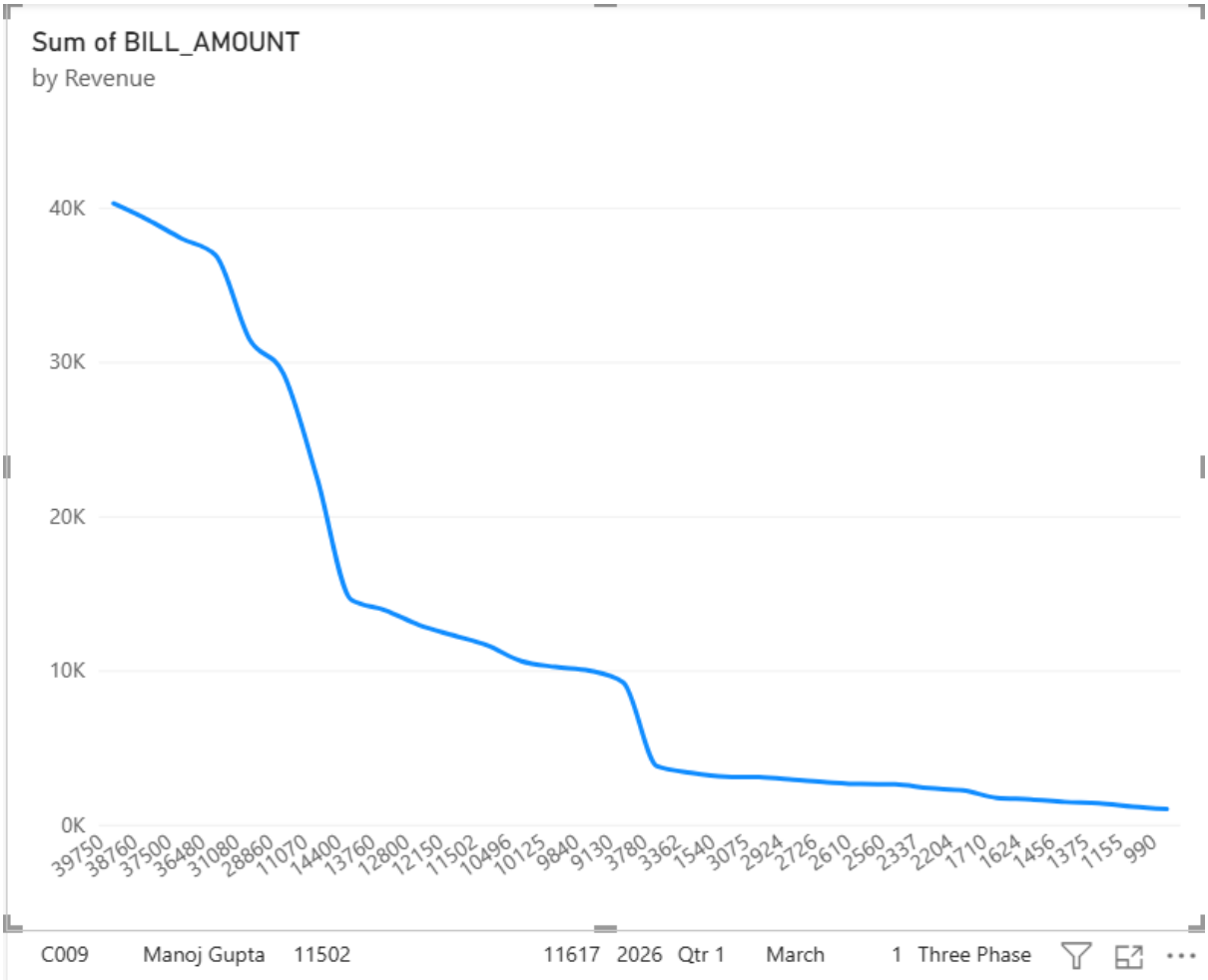

Untitled - Power BI Desktop
File
Home
Insert
Modeling
Design
View
Optimize
Help
Format
Data / Drill
Filters
Visualizations
Data
Page 1
Page 1 of 1


ELECTRICITY_BILLS
BILL_AMOUNT
BILL_ID
BILLING_MONTH
CUSTOMER_ID
RATE_PER_UNIT
Revenue
SURCHARGE
UNITS_CONSUMED
ELECTRICITY_CUSTOME...
Column1
Column2
Column3
Column4
Column5
Column6

Column2	Revenue	Year	Quarter	Month	Day
Amit Shah	14400	2026	Qtr 1	March	1
Ananya Rao	2610	2026	Qtr 1	February	1
Ananya Rao	2726	2026	Qtr 1	March	1
Arjun Kumar	1375	2026	Qtr 1	January	1
Arjun Kumar	1540	2026	Qtr 1	February	1
David Thomas	12800	2026	Qtr 2	April	1
David Thomas	13760	2026	Qtr 2	April	1
Divya Menon	1155	2026	Qtr 1	February	1
Farhan Ali	10496	2026	Qtr 2	April	1
Farhan Ali	11070	2026	Qtr 1	March	1
Gopal Krishna	10125	2026	Qtr 2	April	1
Harish Rao	2924	2026	Qtr 1	March	1
Ishita Roy	1456	2026	Qtr 2	April	1
Kaarthi Raju	2204	2026	Qtr 1	March	1
Kiran Reddy	37500	2026	Qtr 1	January	1
Kiran Reddy	39750	2026	Qtr 1	March	1
Lakshmi Devi	3075	2026	Qtr 1	March	1



C009	Manoj Gupta	11502	11617	2026	Qtr 1	March	1	Three Phase



C009 Manoj Gupta 11502 11617 2026 Qtr 1 March 1 Three Phase